GEOGRAPHY

PHYSICAL GEOGRAPHY

GEOMORPHOLOGY

MODULE - 39

COASTAL LANDFORMS

- EROSIONAL LANDFORMS
- DEPOSITIONAL LANDFORMS

COASTAL LANDFORMS

EROSIONAL LANDFORMS

1 CLIFFS, TERRACES, CAVES AND STACKS

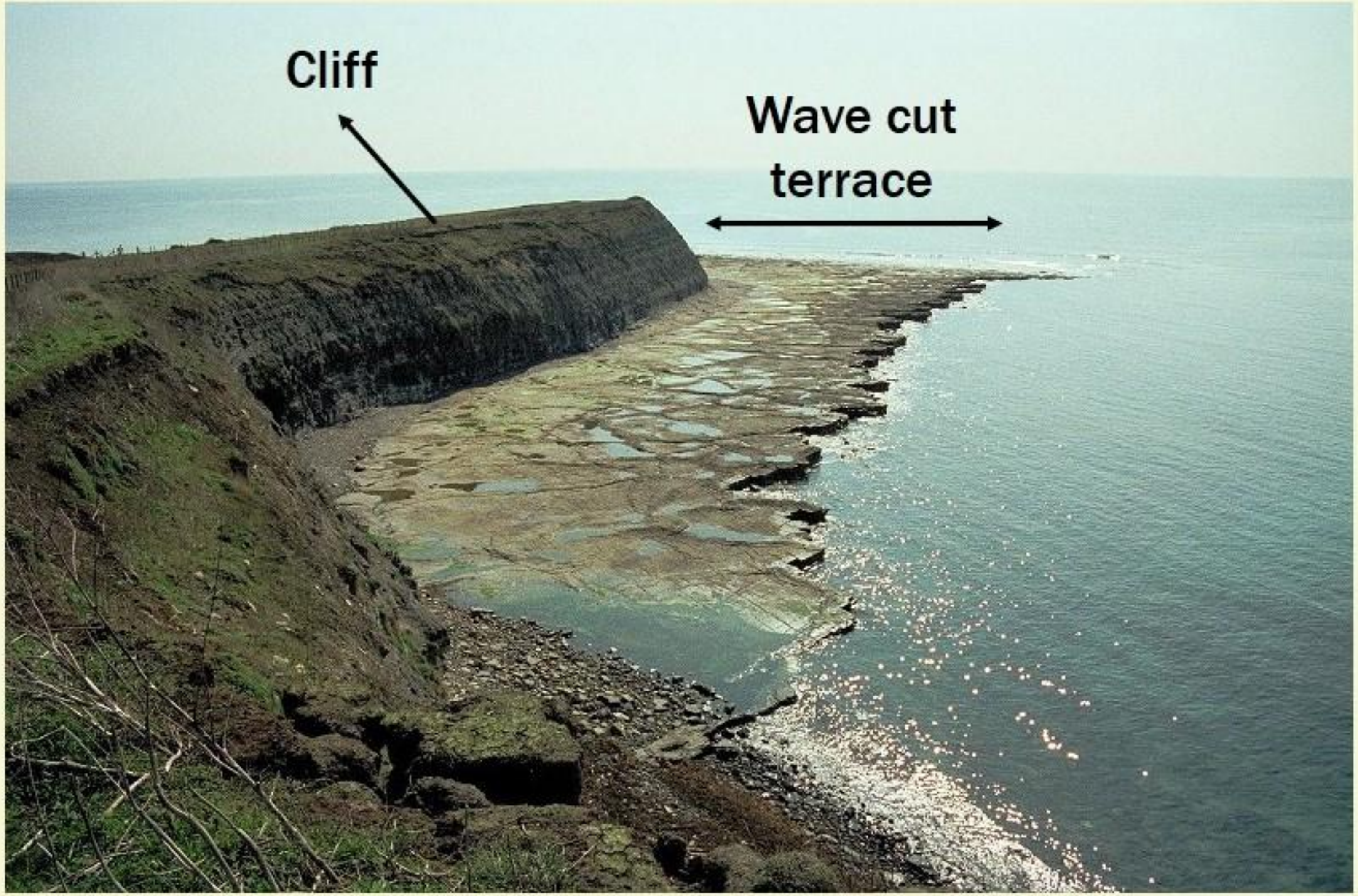
1. Wave-cut cliffs and terraces are two forms usually found where erosion is the dominant shore process.
2. Almost all sea cliffs are steep, and may range from a few metre to 30 m or more.
3. At the foot of such cliffs there may be a flat or gently sloping platform covered by rock debris derived from the sea cliffs behind.
4. Such platforms occurring at elevations above the average height of waves is called a wave-cut terrace.

Learning Space

COASTAL LANDFORMS

EROSIONAL LANDFORMS

1 CLIFFS, TERRACES, CAVES AND STACKS



COASTAL LANDFORMS

EROSIONAL LANDFORMS

1 CLIFFS, TERRACES, CAVES AND STACKS

SEA-STACKS

5. The lashing of waves against the base of the cliff and the rock debris that gets smashed against the cliff along with lashing out waves create hollows and these hollows get widened and deepened to form sea caves.

6. The roofs of caves collapse and the sea cliffs recede further inland. Retreat of the cliff may leave some remnants of rock standing isolated as small islands just off the shore.

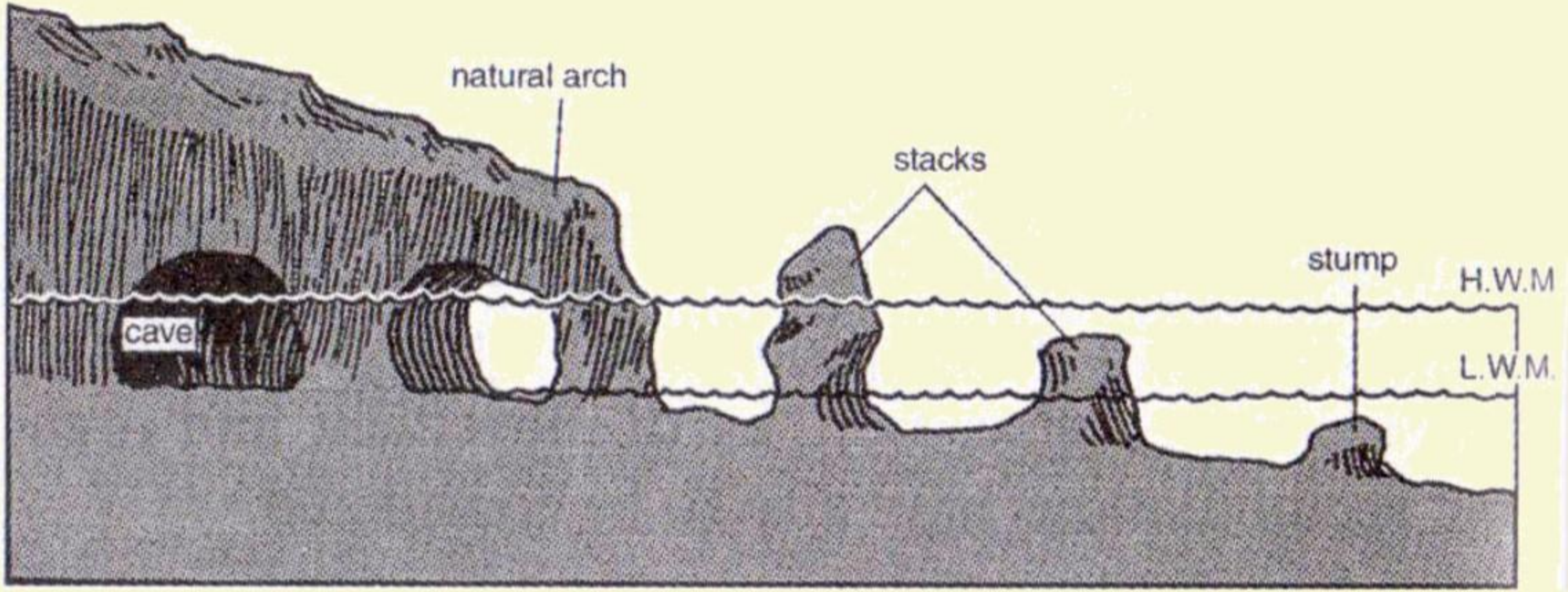
7. Such resistant masses of rock, originally parts of a cliff or hill are called sea-stacks.

Learning Space

COASTAL LANDFORMS

EROSIONAL LANDFORMS

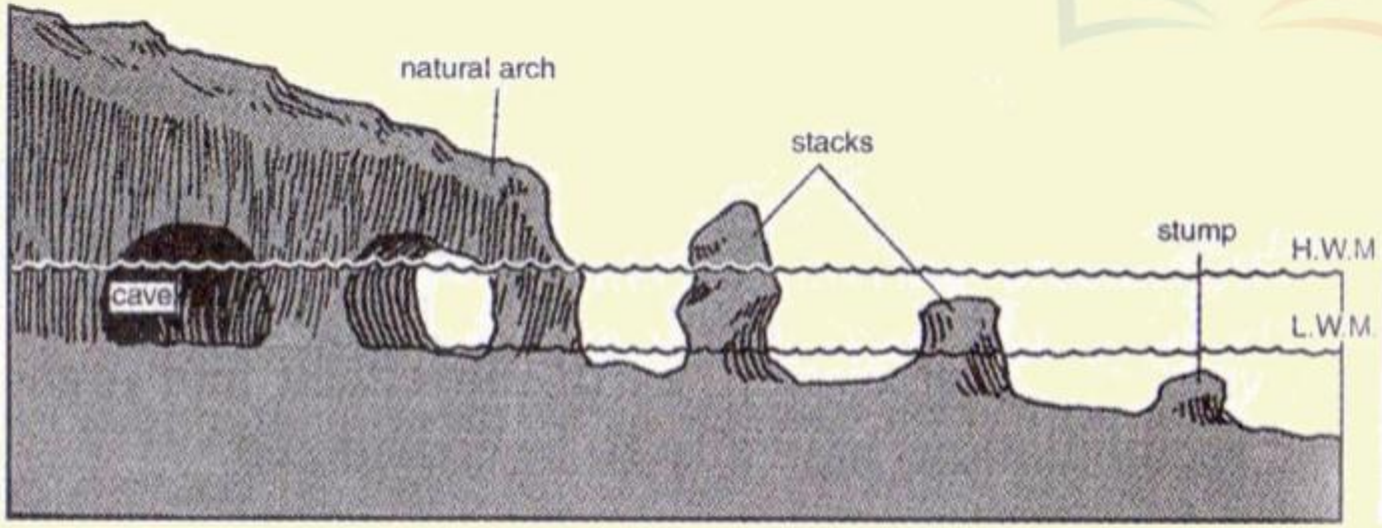
1 CLIFFS, TERRACES, CAVES AND STACKS



COASTAL LANDFORMS

EROSIONAL LANDFORMS

1 CLIFFS, TERRACES, CAVES AND STACKS



8 Prolonged wave attack on the base of a cliff excavates holes in regions of local weakness called caves.

9 Where two caves are eroded on either side of a headland they may eventually join to form a natural arch.

10 Further erosion by waves will ultimately lead to the total collapse of the arch.

11 The seaward portion of the eroded cliff will remain as a pillar of rock known as a stack.

Learning Space

COASTAL LANDFORMS

EROSIONAL LANDFORMS

1 CLIFFS, TERRACES, CAVES AND STACKS

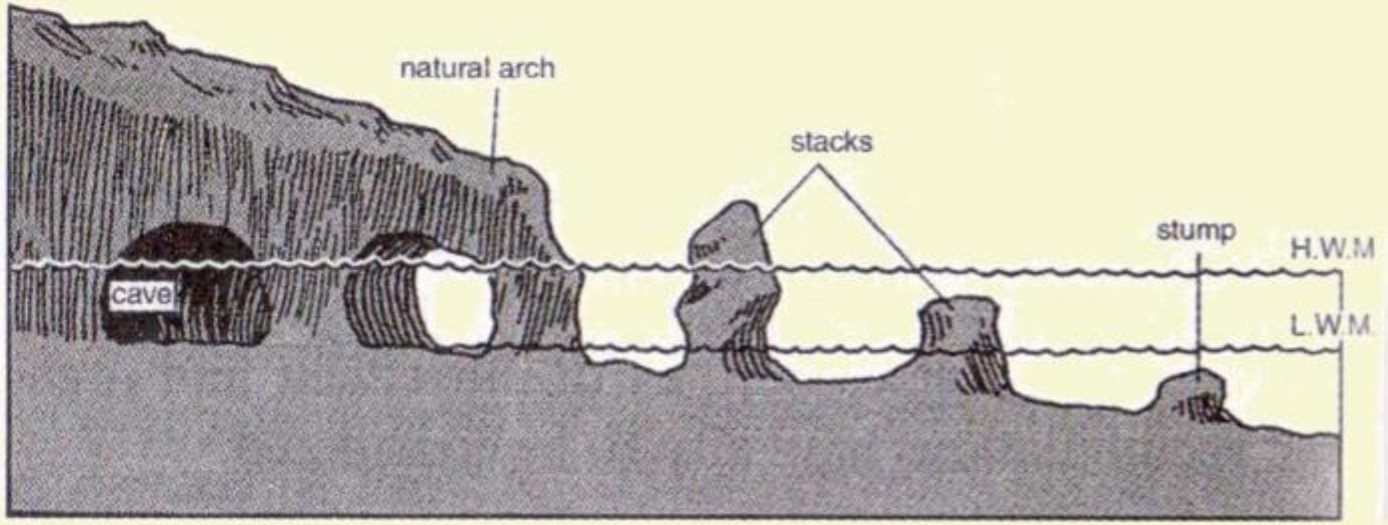
SEA CAVES

NATURAL ARCHES

COASTAL LANDFORMS

EROSIONAL LANDFORMS

1 CLIFFS, TERRACES, CAVES AND STACKS



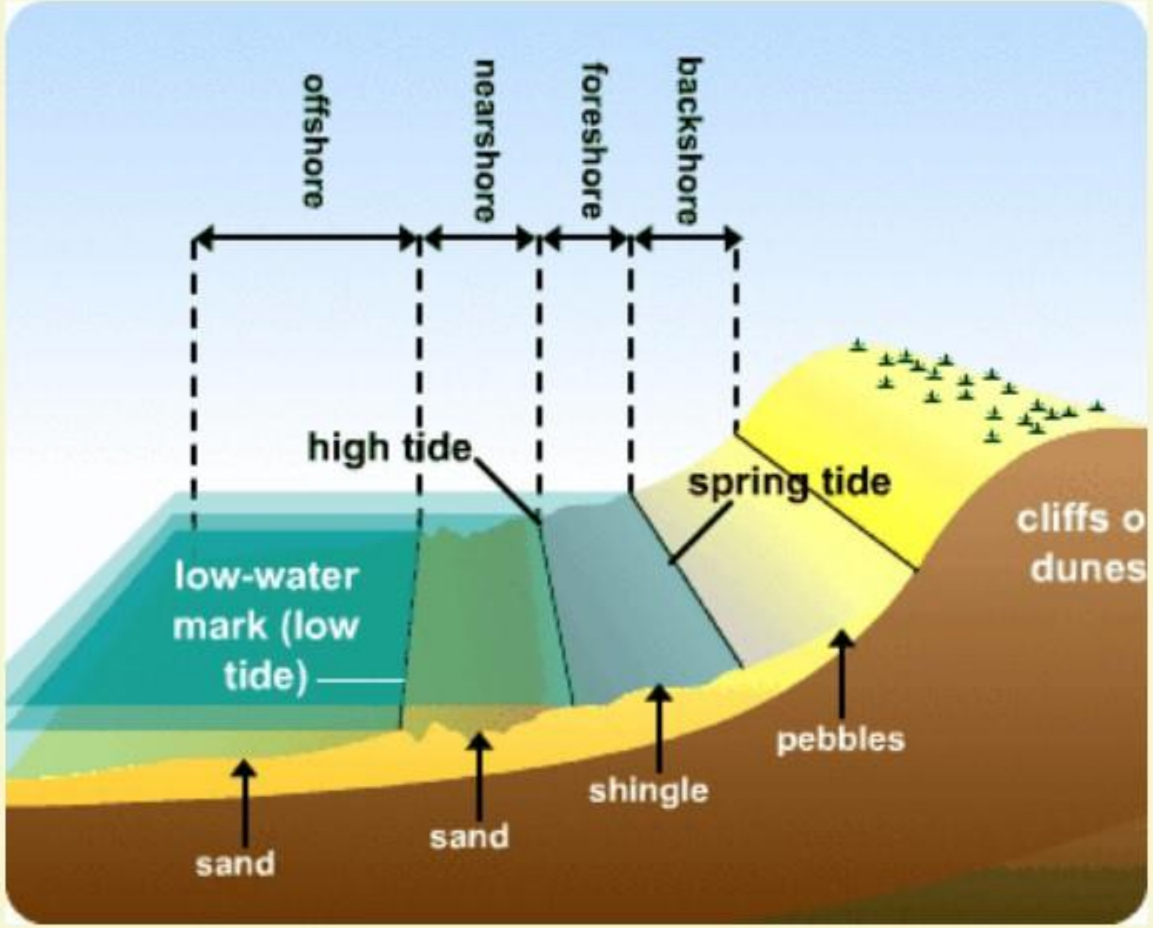
12 In course of time, these stubborn stacks will gradually get removed.

13 The vertical rock pillars are eroded, leaving behind only the stumps which are only just visible just above the sea level.

COASTAL LANDFORMS

EROSIONAL LANDFORMS

1 CLIFFS, TERRACES, CAVES AND STACKS



14 Like all other features, sea stacks are also temporary and eventually, coastal hills and cliffs will disappear because of wave erosion giving rise to narrow coastal plains.

15 With on rush of deposits from over the land behind, these may get covered up by alluvium or may get covered up by shingle or sand to form a wide beach.

COASTAL LANDFORMS

DEPOSITIONAL LANDFORMS

1 BEACHES AND DUNES

BEACHES

1. Beaches are characteristic of shorelines that are dominated by deposition. But, may occur as patches along even the rugged shores.
2. Most of the sediment making up the beaches comes from land carried by the streams and rivers or from wave erosion.
3. Beaches are temporary features.
4. The sandy beach which appears to permanent may be reduced to a very narrow strip of coarse pebbles in some other season.

Learning Space

COASTAL LANDFORMS

DEPOSITIONAL LANDFORMS

1 BEACHES AND DUNES

SHINGLES IN BEACHES

5 Most of the beaches are made up of sand sized materials.

6 Beaches called shingle beaches contain excessively small pebbles and even cobbles.

COASTAL LANDFORMS

DEPOSITIONAL LANDFORMS

1 BEACHES AND DUNES

7 Just behind the beach, the sands lifted and winnowed from over the beach surfaces will be deposited as sand dunes.

8 Sand dunes forming long ridges parallel to the coastline are very common along low sedimentary coasts.

COASTAL LANDFORMS

DEPOSITIONAL LANDFORMS

2 BARS, BARRIERS & SPITS

OFF-SHORE BAR

1. A ridge of sand and shingle formed in the sea in the off-shore zone lying approximately parallel to the coast is called an off-shore bar.
2. An off-shore bar which is exposed due to further addition of sand is termed a barrier–bar.

COASTAL LANDFORMS

DEPOSITIONAL LANDFORMS

2 BARS, BARRIERS & SPITS

SPITS

3 The off-shore bars and barriers commonly form across the mouth of the river or at the entrance of a bay.

4 Sometimes such barrier bars get keyed up to one end of the bay and then they are called spits.

5 Spits may also develop attached to headlands/hills.

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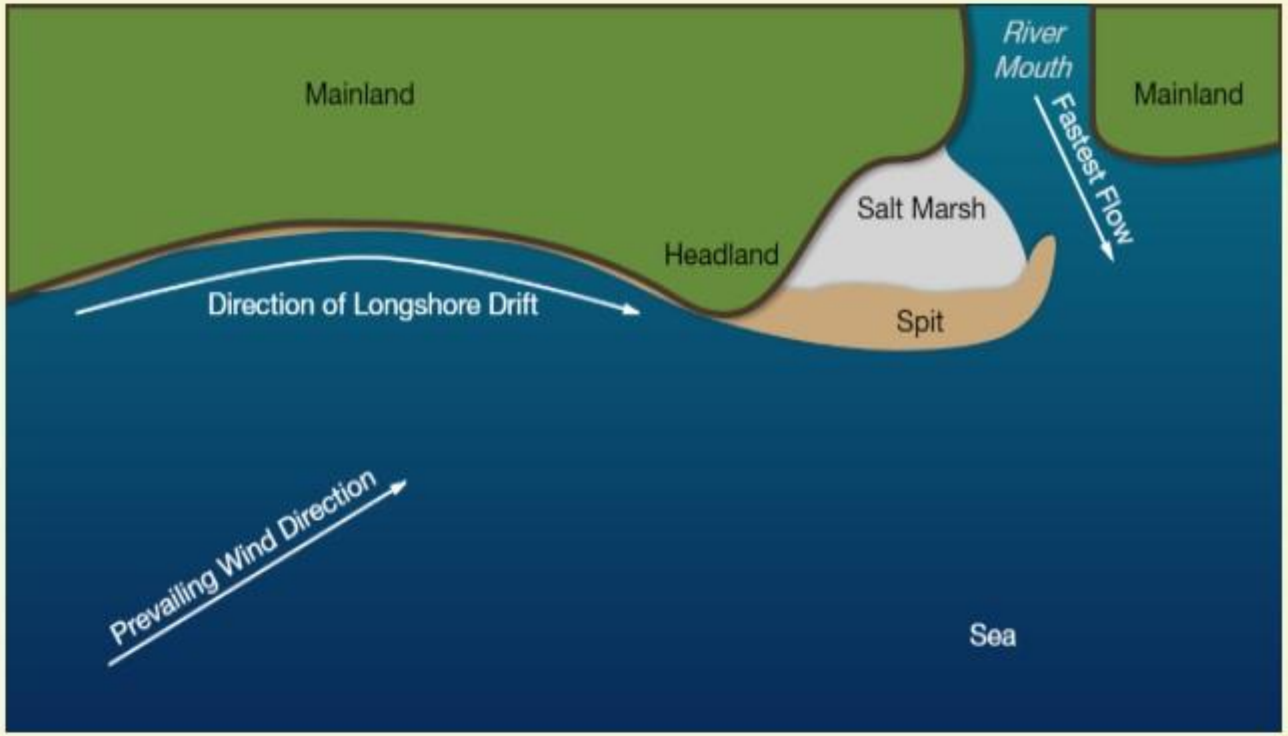
COASTAL LANDFORMS

DEPOSITIONAL LANDFORMS

2 BARS, BARRIERS & SPITS



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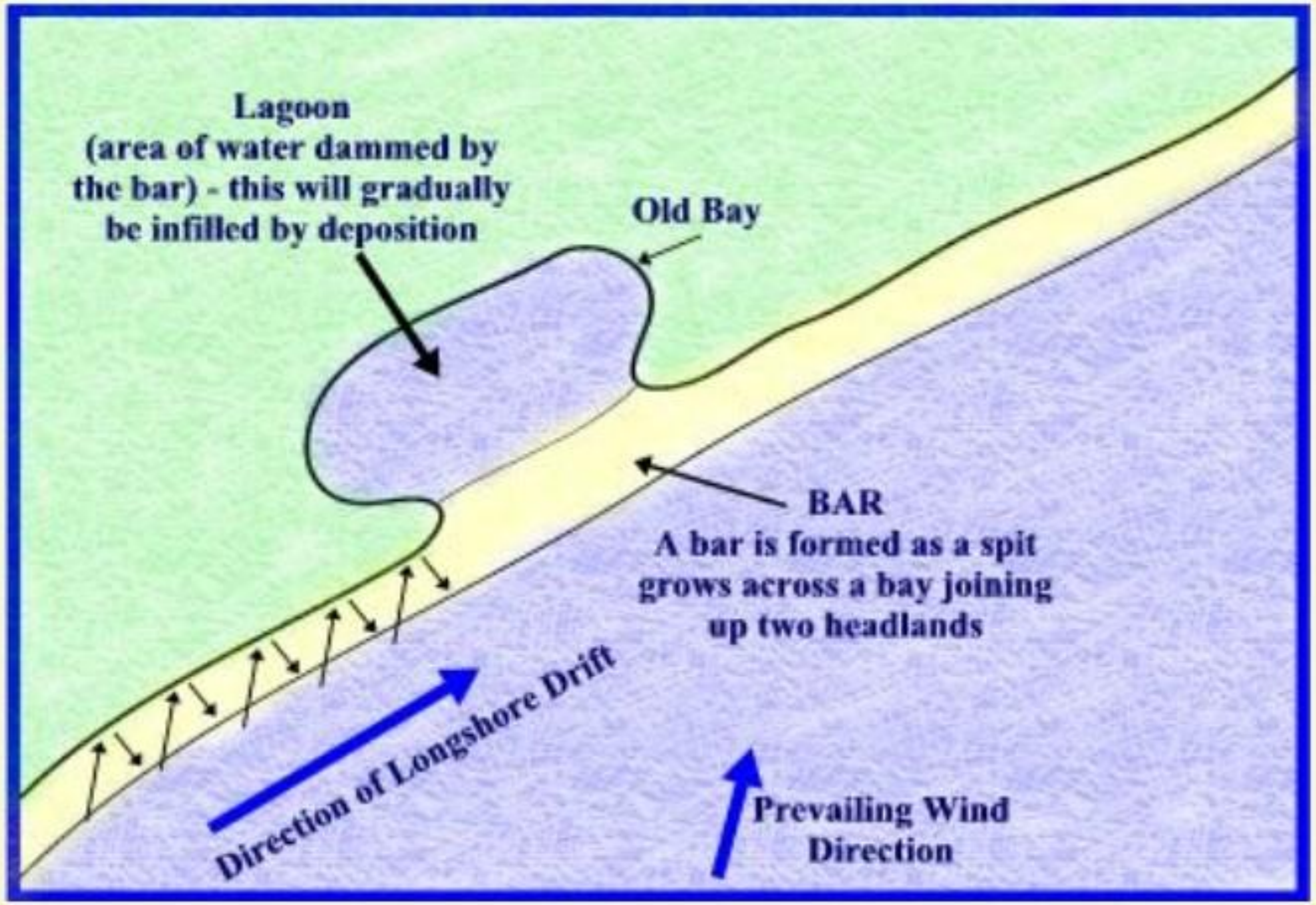
DEPOSITIONAL LANDFORMS

2 BARS, BARRIERS & SPITS



6 The barriers, bars and spits at the mouth of the bay gradually extend leaving only a small opening of the bay into the sea and the bay will eventually develop into a lagoon.

7 The lagoons get filled up gradually by sediment coming from the land or from the beach itself (aided by wind) and a broad and wide coastal plain may develop replacing lagoon.

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COASTAL LANDFORMS

DO YOU KNOW?

- *The coastal off-shore bars offer the first buffer against storm or tsunami by absorbing most of their destructive force.*
- *Then come the barriers, beaches, beach dunes and mangroves, if any, to absorb the destructive force of storm and tsunami waves.*
- *So, if we do anything which disturbs the 'sediment budget' and the mangroves along the coast, these coastal forms will get eroded away leaving human habitations to bear first strike of storm and tsunami waves.*

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